

SNN Surrogate Gradient **tri** β Parameter Comparison Robustness Sweep under Data Heterogeneity ($f \leq 1$)

ByzFL SNN Benchmark

July 2, 2026

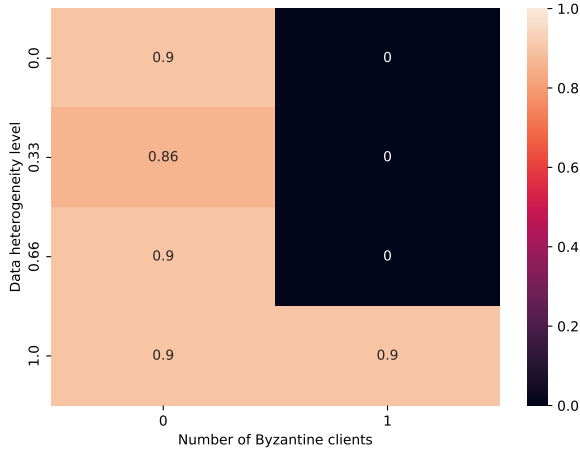
1 Introduction

This document presents a comparison of the Spiking Neural Network (SNN) robustness on the MNIST dataset using different scaling factors β for the Triangular (**tri**) surrogate gradient. The evaluation spans:

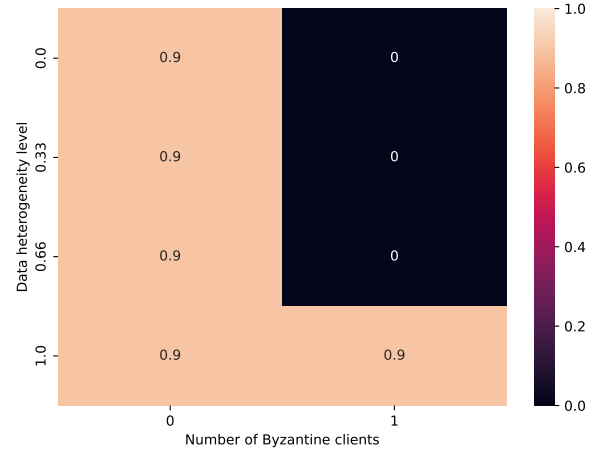
- **Surrogate Gradient Parameter:** $\beta \in \{0.5, 0.75, 1.0, 1.25, 1.5\}$
- **Byzantine Nodes:** $f \in \{0, 1\}$ out of $N = 10$ honest nodes
- **Data Heterogeneity:** Non-IID levels $\gamma \in \{1.0, 0.66, 0.33, 0.0\}$
- **Attacks:** Worst-case outcome across SignFlipping and Optimal A Little Is Enough (**Optimal_ALittleIsEnough_neg1**)
- **Aggregators:** Best aggregated result across Centered Clipping and Geometric Median (pre-aggregated with NNM and ARC).

2 Aggregated Test Accuracy Heatmaps

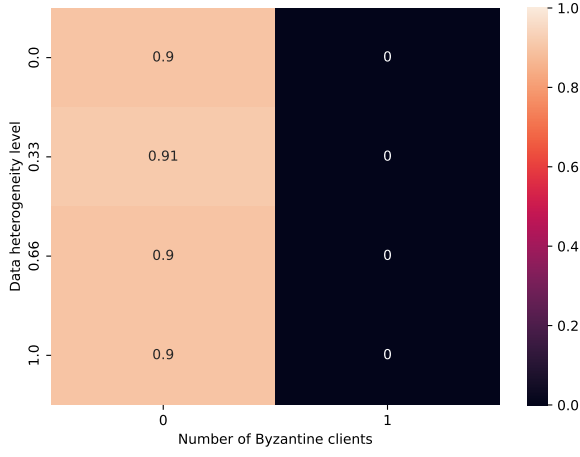
The figure below displays the aggregated test accuracy heatmaps for each tested β parameter value for $f \leq 1$.



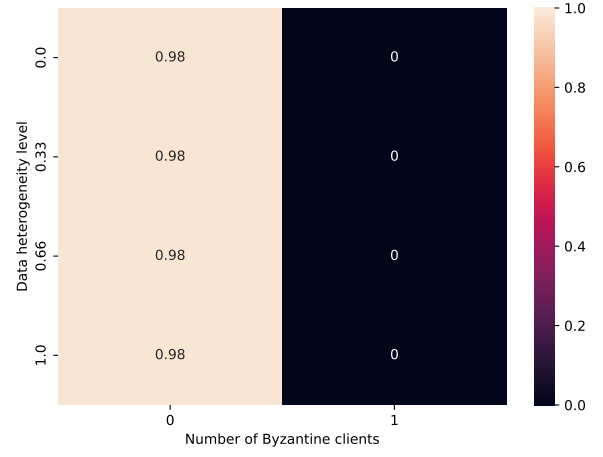
(a) $\beta = 0.5$



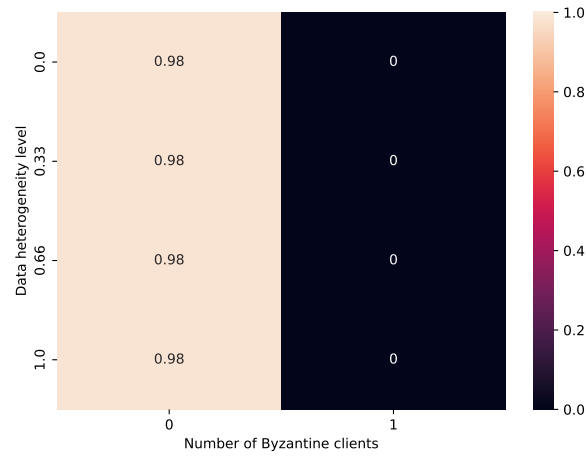
(b) $\beta = 0.75$



(c) $\beta = 1.0$



(d) $\beta = 1.25$



(e) $\beta = 1.5$

Figure 1: Aggregated Test Accuracy Heatmaps across different surrogate gradient β values for SNN ($f \leq 1$).

3 Quantitative Performance Summary

Table 1 compares the overall performance metrics for each β parameter value. The metrics include the average accuracy across all 8 cells in the grid, and the worst-case accuracy.

Table 1: Performance Summary for Different Surrogate Gradient scaling values (β).

Surrogate Scaling (β)	Average Accuracy	Worst-case Accuracy
$\beta = 0.5$	55.62%	0.00%
$\beta = 0.75$	56.03%	0.00%
$\beta = 1.0$	45.02%	0.00%
$\beta = 1.25$	49.06%	0.00%
$\beta = 1.5$	48.95%	0.00%

4 Detailed Grid Tables for each β

This section lists the exact aggregated test accuracies for each data heterogeneity level (γ) and number of Byzantine clients ($f \in \{0, 1\}$).

4.1 Surrogate scaling $\beta = 0.5$

	f = 0	f = 1
$\gamma = 1.0$	89.77%	89.83%
$\gamma = 0.66$	89.82%	0.00%
$\gamma = 0.33$	85.70%	0.00%
$\gamma = 0.0$	89.82%	0.00%

4.2 Surrogate scaling $\beta = 0.75$

	f = 0	f = 1
$\gamma = 1.0$	89.69%	89.57%
$\gamma = 0.66$	89.69%	0.00%
$\gamma = 0.33$	89.69%	0.00%
$\gamma = 0.0$	89.62%	0.00%

4.3 Surrogate scaling $\beta = 1.0$

4.4 Surrogate scaling $\beta = 1.25$

4.5 Surrogate scaling $\beta = 1.5$

	f = 0	f = 1
$\gamma = 1.0$	89.62%	0.00%
$\gamma = 0.66$	89.63%	0.00%
$\gamma = 0.33$	91.25%	0.00%
$\gamma = 0.0$	89.64%	0.00%

	f = 0	f = 1
$\gamma = 1.0$	98.12%	0.00%
$\gamma = 0.66$	98.13%	0.00%
$\gamma = 0.33$	98.12%	0.00%
$\gamma = 0.0$	98.10%	0.00%

	f = 0	f = 1
$\gamma = 1.0$	97.90%	0.00%
$\gamma = 0.66$	97.89%	0.00%
$\gamma = 0.33$	97.92%	0.00%
$\gamma = 0.0$	97.87%	0.00%